IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- **Power/circle-equation reformulation:** rewrote $OM = ON$ as a linear relation between the coefficients of the circumcircle of AKL.
- **Angle parametrization and sine law:** introduced $\alpha, \beta, \gamma, x, y, z$ and derived exact sine-law expressions for BK and CL.
- **Normalized-length quadratics:** set $R = 2BK/AB$, $D = 2CL/AC$ and eliminated y, z, obtaining two compact quadratic relations through a common polynomial F.
- **Oblique coordinates/full elimination:** verified the identity but initially produced unwieldy algebra; this route also exposed and corrected one false intermediate symbolic claim.

- **Cartesian circle computation:** used simple ray-direction coordinates for K, L. Sequential exact polynomial division yielded a sparse certificate expressing the target in terms of the two geometric quadratic relations and the identities $\sin^2 + \cos^2 = 1$.
- **Synthetic/power exploration:** considered equal powers and inversion, but no comparably clear synthetic construction was found.

Current best

A complete analytic proof has been obtained. With $S = \sin\alpha, C = \cos\alpha, U = \sin x, X = \cos x, V = \sin(\alpha + x)$, $R = 2BK/AB$, and $D = 2CL/AC$, the angle conditions imply

$$2U \sin\beta = \sin\gamma\, F(R), \qquad 2U \sin\gamma = \sin\beta\, F(D), \quad F(T) = V(T^2 - 2XT + 2) - ST.$$

A verified explicit polynomial identity then gives

$$b(CP + SQ) - cP = \frac{b^2 - c^2}{2}$$

for the Cartesian coefficients of the circumcircle of AKL, which is exactly $OM^2 - ON^2 = 0$. The sparse certificate and its scaling were independently checked by `code/verify_sparse_identity.py`; the ray orders, positivity of every divisor, coordinate directions, and all circle formulas have also been audited in `scratch/audit.md`.

Full proof

Let

$$\alpha = \angle BAC, \qquad \beta = \angle ABC, \qquad \gamma = \angle ACB,$$

and set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK.$$

Also write

$$a = BC, \qquad b = CA, \qquad c = AB.$$

Because K lies inside $\angle LBA$, the rays at B occur in the order

$$BA,\ BK,\ BL,\ BC;$$

because L lies inside $\angle ACK$, the rays at C occur in the order

$$CA,\ CL,\ CK,\ CB.$$

In particular, all angles and sines used below have the indicated positive values.

1. Relations forced by the angle conditions

In $\triangle BMK$ we have $BM = c/2$, $\angle MBK = x$, and $\angle BMK = z$. Hence

$$BK = \frac{c}{2}\frac{\sin z}{\sin(x+z)}. \qquad (1)$$

Moreover,

$$\angle KBC = \beta - x, \qquad \angle BCK = \gamma - x - z,$$

so $\angle BKC = \alpha + 2x + z$. Thus

$$BK = a\frac{\sin(\gamma - x - z)}{\sin(\alpha + 2x + z)}. \qquad (2)$$

Similarly, sine rule in $\triangle CNL$ and $\triangle BCL$ gives

$$CL = \frac{b}{2}\frac{\sin y}{\sin(x+y)} = a\frac{\sin(\beta - x - y)}{\sin(\alpha + 2x + y)}. \qquad (3)$$

Normalize the circumdiameter of ABC to 1. Then

$$a = \sin\alpha, \qquad b = \sin\beta, \qquad c = \sin\gamma. \qquad (4)$$

Put

$$S = \sin\alpha, \quad C = \cos\alpha, \quad U = \sin x, \quad X = \cos x, \quad V = \sin(\alpha + x) = SX + CU, \qquad (5)$$

and

$$R = \frac{2BK}{c}, \qquad D = \frac{2CL}{b}. \qquad (6)$$

Thus

$$R = \frac{\sin z}{\sin(x+z)}, \qquad D = \frac{\sin y}{\sin(x+y)}. \qquad (7)$$

Let (θ, t, ρ) denote either (γ, z, R) or (β, y, D). Equating the appropriate expressions in (1)–(3) and using (4) gives

$$\sin\theta \sin t \sin(\alpha + 2x + t) = 2S\sin(x+t)\sin(\theta - x - t). \qquad (8)$$

Since $t > 0$, divide by $\sin^2 t$. From $\rho = \sin t / \sin(x+t)$,

$$\cot t = \frac{1/\rho - X}{U}. \qquad (9)$$

Writing all terms in (8) in terms of $\cot t$ and using (9) yields

$$-\rho^2 \sin\theta\, U \sin(\alpha + x) + \rho\sin\theta(\cos(\alpha - x) - X\cos(\alpha + 2x)) - 2SU\sin(\theta - x) = 0. \qquad (10)$$

We use

$$\cos(\alpha - x) - X\cos(\alpha + 2x) = U(2CUX - 2SU^2 + 3S), \qquad (11)$$

and

$$\cot\gamma = \frac{b - Cc}{Sc}, \qquad \cot\beta = \frac{c - Cb}{Sb}. \qquad (12)$$

Dividing (10) by $U\sin\theta$ and applying (11)–(12) gives

$$2Ub = cF(R), \qquad 2Uc = bF(D), \qquad (13)$$

where

$$F(T) = V(T^2 - 2XT + 2) - ST. \qquad (14)$$

Indeed, using $V = SX + CU$ and $U^2 + X^2 = 1$,

$$F(T) = VT^2 - (2CUX - 2SU^2 + 3S)T + 2SX + 2CU,$$

which is exactly the expression obtained from (10).

2. The circle computation

Place

$$A = (0, 0), \qquad B = (c, 0), \qquad C = (bC, bS).$$

Put $W = \cos(\alpha + x) = CX - SU$. The ray BK has direction $(-X, U)$ and the ray CL has direction $(-W, -V)$, so

$$K = \left(c - \frac{cRX}{2}, \frac{cRU}{2}\right), \qquad (15)$$

$$L = \left(bC - \frac{bDW}{2}, bS - \frac{bDV}{2}\right). \qquad (16)$$

Consequently,

$$|K|^2 = c^2\left(1 - RX + \frac{R^2}{4}\right), \qquad |L|^2 = b^2\left(1 - DX + \frac{D^2}{4}\right). \qquad (17)$$

Write the circumcircle of AKL as

$$\xi^2 + \eta^2 - P\xi - Q\eta = 0. \qquad (18)$$

Its centre is $O = (P/2, Q/2)$. Set

$$\Delta = K_x L_y - K_y L_x.$$

Since the circumcentre is defined, A, K, L are noncollinear, hence $\Delta \neq 0$. Cramer's rule gives

$$P = \frac{P_0}{\Delta}, \qquad Q = \frac{Q_0}{\Delta}, \qquad (19)$$

where

$$P_0 = |K|^2 L_y - K_y |L|^2, \qquad Q_0 = K_x |L|^2 - |K|^2 L_x. \qquad (20)$$

Define

$$\mathcal{T} = 2(b(CP_0 + SQ_0) - cP_0) - (b^2 - c^2)\Delta. \qquad (21)$$

Let

$$E_R = 2Ub - cF(R), \qquad E_D = 2Uc - bF(D).$$

Substituting (14)–(17) and (20)–(21), and collecting terms, gives the polynomial identity

$$4V\mathcal{T} = A_0 E_R + B_0 E_D + (C^2 + S^2 - 1)G_0 + (X^2 + U^2 - 1)H_0, \qquad (22)$$

where

$$A_0 = bc(C^2DUb - CDUc + DS^2Ub - DSXc + 2Sc),$$
$$B_0 = bc(CRUb + RSXb - RUc - 2Sb),$$
$$G_0 = Ubc^2(2CDRUXb - 2CDUb + 2DRSX^2b - DRSb - 2DSXb - 2RUc),$$
$$H_0 = Sbc(CDRUb^2 - CDRUc^2 + DRSXb^2 - DRSXc^2 - 2DSb^2 + 2RSc^2).$$

Identity (22) is verified by direct expansion of the displayed expressions.

By (13), $E_R = E_D = 0$, and $C^2 + S^2 = X^2 + U^2 = 1$. Also $V = \sin(\alpha + x) > 0$: indeed $0 < x < \gamma$, so $0 < \alpha + x < \pi$. Hence (22) gives $\mathcal{T} = 0$. By (19) and (21),

$$b(CP + SQ) - cP = \frac{b^2 - c^2}{2}. \qquad (23)$$

Finally,

$$M = \left(\frac{c}{2}, 0\right), \qquad N = \left(\frac{bC}{2}, \frac{bS}{2}\right).$$

Since $O = (P/2, Q/2)$,

$$\begin{aligned} OM^2 - ON^2 &= -\frac{cP}{2} + \frac{b(CP + SQ)}{2} + \frac{c^2 - b^2}{4} \\ &= 0 \end{aligned}$$

by (23). Therefore $OM = ON$.